13. Intersections of general sets

In this chapter we study the following integralgeometric question. Let A and B be Borel sets in $\mathbf{R}^n$. What kind of relations are there between the Hausdorff dimensions of A, B and $A \cap fB$ when f runs through the isometries of $\mathbf{R}^n$? What one could hope for is that often

$$\dim(A \cap fB) = \dim A + \dim B - n \tag{13.1}$$

provided the right hand side is non-negative. In the case where B is an m-plane such results were already established in Chapter 10. Those methods could be generalized to the case where B is a sufficiently nice m-dimensional surface, for example a C^1 submanifold.

Intersection measures and energies

As in Chapter 10, we shall use here capacities and energy-integrals. Starting from Radon measures μ and ν with supports in A and B, respectively, we attempt to construct measures $\mu \cap f_\sharp \nu$ with supports in $A \cap fB$ such that if $I_s(\mu) < \infty$, $I_t(\nu) < \infty$ and $s + t - n > 0$, then $I_{s+t-n}(\mu \cap f_\sharp \nu) < \infty$ for almost all f. Unfortunately, we can do this only under the additional assumption that either s or t is bigger than $(n+1)/2$. This will lead to the inequality "$\geq$" in (13.1) for isometries f in a set of positive measure provided either $\dim A$ or $\dim B$ is bigger than $(n+1)/2$. (Of course, one cannot hope this for almost all f, since the intersection may be empty very often.) The necessity of the $((n+1)/2)$-assumption is unknown. The opposite inequality "$\leq$" in (13.1) can be false for all isometries f, but we shall show that under some additional hypotheses, it holds. For integralgeometric formulas relating the measures of A, B and $A \cap fB$ in the case where both A and B have some regularity properties, i.e. they are rectifiable, see e.g. Federer [3, 3.2.48].

Recall that $f \colon \mathbf{R}^n \to \mathbf{R}^n$ is an isometry if it preserves distances: $|f(x) - f(y)| = |x - y|$ for all $x, y \in \mathbf{R}^n$. This is equivalent to saying that

$$f = \tau_z \circ g \quad \text{for some } z \in \mathbf{R}^n,\ g \in O(n)$$

(recall the notations from 3.15). In order to construct the measures $\mu \cap (\tau_z \circ g)_\sharp \nu$ we shall slice as in 10.1 the product measure $\mu \times g_\sharp \nu$ by the n-planes

$$V_z = \{(x, y) \in \mathbf{R}^n \times \mathbf{R}^n : x = y + z\}, \quad z \in \mathbf{R}^n,$$

parallel to the diagonal $W = \{(x,y) : x = y\}$ of $\mathbf{R}^n \times \mathbf{R}^n$. The slices thus obtained are then projected to $\mathbf{R}^n$ by π, $\pi(x,y) = x$. The reason that this gives the desired measures is the simple formula

$$A \cap (\tau_z \circ g)B = \pi\big[(A \times (gB)) \cap V_z\big] \tag{13.2}$$

for the intersection of A and $(\tau_z \circ g)B$. This method is from Mattila [7] and it is the same as used for the construction of intersection currents in Federer [3, 4.3.20].

13.3. The intersection measures. Let μ and ν be Radon measures on $\mathbf{R}^n$ with compact support, and let $g \in O(n)$. The orthogonal complement of $W = \{(x,y) : x = y\}$ is $W^\perp = \{(x,y) : x = -y\}$. Thus by (10.3) there exists for $\mathcal{L}^n$ almost all $z \in \mathbf{R}^n$ a Radon measure $(\mu \times g_\sharp \nu)_{W,(z,-z)/2}$ such that for $\psi \in C_0^+(\mathbf{R}^n \times \mathbf{R}^n)$,

$$\begin{aligned}
&\int \psi \, d(\mu \times g_\sharp \nu)_{W,(z,-z)/2} = \lim_{\delta \downarrow 0} (2\delta)^{-n} \int_{W_{(z,-z)/2}(\delta)} \psi \, d(\mu \times g_\sharp \nu) \\
&= \lim_{\delta \downarrow 0} (2\delta)^{-n} \iint_{\{(x,y) : |x - gy - z|/\sqrt{2} \le \delta\}} \psi(x, gy) \, d\mu x \, d\nu y,
\end{aligned}$$

since $P_{W^\perp}(x,y) = (x - y, y - x)/2$, and so

$$\begin{aligned}
W_{(z,-z)/2}(\delta) &= \big\{(x,y) : d((x,y), W_{(z,-z)/2}) \le \delta\big\} \\
&= \big\{(x,y) : |P_{W^\perp}((x,y) - (z,-z)/2)| \le \delta\big\} \\
&= \big\{(x,y) : |\tfrac{1}{2}(x - y - z, y - x + z)| \le \delta\big\} \\
&= \big\{(x,y) : |x - y - z|/\sqrt{2} \le \delta\big\}.
\end{aligned}$$

We define

$$\mu \cap (\tau_z \circ g)_\sharp \nu = 2^{n/2} \alpha(n)^{-1} \pi_\sharp \big[(\mu \times g_\sharp \nu)_{W,(z,-z)/2}\big],$$

where $\pi(x,y) = x$. Then for $\varphi \in C_0^+(\mathbf{R}^n)$

$$\begin{aligned}
&\int \varphi \, d\mu \cap (\tau_z \circ g)_\sharp \nu \\
&= 2^{n/2} \alpha(n)^{-1} \int \varphi(\pi(x,y)) \, d(\mu \times g_\sharp \nu)_{W,(z,-z)/2}(x,y) \\
&= \lim_{\delta \downarrow 0} \alpha(n)^{-1} (\delta\sqrt{2})^{-n} \iint_{\{(x,y) : |x - gy - z|/\sqrt{2} \le \delta\}} \varphi(x) \, d\mu x \, d\nu y \\
&= \lim_{\delta \downarrow 0} \alpha(n)^{-1} \delta^{-n} \iint_{\{(x,y) : |x - gy - z| \le \delta\}} \varphi(x) \, d\mu x \, d\nu y.
\end{aligned}$$

For $g \in O(n)$ and $z \in \mathbf{R}^n$ define a continuous map $S_g\colon \mathbf{R}^n \times \mathbf{R}^n \to \mathbf{R}^n$ by

$$S_g(x,y) = x - gy$$

and define

$$W_{g,z}(\delta) = \big\{(x,y) \in \mathbf{R}^n \times \mathbf{R}^n : |S_g(x,y) - z| \leq \delta\big\}.$$

Then we have

$$(13.4)\quad \int \varphi\, d\mu \cap (\tau_z \circ g)_\sharp \nu = \lim_{\delta \downarrow 0} \alpha(n)^{-1}\delta^{-n} \int_{W_{g,z}(\delta)} \varphi(x)\, d(\mu \times \nu)(x,y)$$

for $\varphi \in C_0^+(\mathbf{R}^n)$. The properties (10.4–6) of the measures $\mu_{W,a}$ in Chapter 10 now turn immediately into the following properties for the intersection measures.

$$(13.5)\qquad \operatorname{spt} \mu \cap (\tau_z \circ g)_\sharp \nu \subset \operatorname{spt} \mu \cap (\tau_z \circ g)(\operatorname{spt} \nu).$$

Let h be a lower semicontinuous non-negative function on $\mathbf{R}^n$. Then

$$(13.6)\qquad \begin{aligned} &\int h\, d\mu \cap (\tau_z \circ g)_\sharp \nu \\ &\leq \liminf_{\delta \downarrow 0} \alpha(n)^{-1}\delta^{-n} \int_{W_{g,z}(\delta)} h(x)\, d(\mu \times \nu)(x,y), \end{aligned}$$

and

$$(13.7)\qquad \int_B \int h\, d\mu \cap (\tau_z \circ g)_\sharp \nu\, d\mathcal{L}^n z \leq \int_{S_g^{-1}(B)} h(x)\, d(\mu \times \nu)(x,y)$$

for any Borel set $B \subset \mathbf{R}^n$. Equality holds in (13.7) if $S_{g\sharp}(\mu \times \nu) \ll \mathcal{L}^n$. Note that the application of (10.6) involves a change of variable $z \mapsto a = (z,-z)/2$ under which $d\mathcal{L}^n z$ transforms to $2^{-n/2}\alpha(n) d(\mathcal{H}^n \llcorner W^\perp)a$. Alternatively one can apply the differentiation theorem 2.12 to $S_{g\sharp}(\mu \times \nu)$ without going via (10.6). Also the whole construction of the intersection measures can be performed directly in this way.

In order to use these measures we need analogues of Theorems 9.7 and 10.7. Their proofs are based on the following modification of Lemma 12.13.

13.8. Lemma. *If $(n+1)/2 \le s < n$, $0 < \delta \le r < \infty$, and μ is a Radon measure on $\mathbf{R}^n$ with compact support, then*

$$\mu \times \mu\big(\{(x,y) : r \le |x-y| \le r+\delta\}\big) \le cI_s(\mu)\,\delta r^{s-1}$$

where c is a constant depending only on n and s.

Proof. Let g be the characteristic function of the annulus $\{z \in \mathbf{R}^n : r < |z| < r+\delta\}$. By (12.8) we have again

$$\widehat{g}(x) = c_1|x|^{-n} \int_{r|x|}^{(r+\delta)|x|} J_{(n-2)/2}(u)\,u^{n/2}\,du.$$

Here and below the constants $c_1, \dots, c_4$ will depend only on n. If $r|x| \ge 1$, we infer from this and (12.5)

$$\begin{aligned}\big|\widehat{g}(x)\big| &\le c_2|x|^{-n}((r+\delta)|x|)^{n/2-1/2}\,\delta|x| \\ &\le c_2 2^{(n-1)/2}\,\delta(r|x|)^{(n+1)/2-s} r^{s-1}|x|^{s-n} \\ &\le c_2 2^{(n-1)/2}\,\delta r^{s-1}|x|^{s-n}.\end{aligned}$$

If $r|x| \le 1$, we use (12.6) to get a similar estimate:

$$\begin{aligned}\big|\widehat{g}(x)\big| &\le c_3|x|^{-n}((r+\delta)|x|)^{n-1}\,\delta|x| \\ &\le c_3 2^{n-1}\,\delta\big(r|x|\big)^{n-s} r^{s-1}|x|^{s-n} \\ &\le c_3 2^{n-1}\,\delta r^{s-1}|x|^{s-n}.\end{aligned}$$

Using these we compute as in the proof of Lemma 12.13

$$\begin{aligned}&\mu \times \mu\big(\{(x,y) : r < |x-y| < r+\delta\}\big) \\ &\le (2\pi)^{-n} \int |\widehat{g}|\,|\widehat{\mu}|^2\,d\mathcal{L}^n \\ &\le c_4\delta r^{s-1} \int |x|^{s-n}\big|\widehat{\mu}(x)\big|^2\,dx = c\delta r^{s-1} I_s(\mu),\end{aligned}$$

by Lemma 12.12, which clearly yields the lemma. □

The above lemma does not hold for any $s < (n+1)/2$ as was shown in Mattila [9]. However the necessity of the assumption $(n+1)/2 \le t$ in the applications below is not known.

13.9. Lemma. *Suppose $0 < s < n$, $0 < t < n$, $s + t \geq n$, and $t \geq (n+1)/2$. If μ and ν are Radon measures on $\mathbf{R}^n$ with compact support and if $I_s(\mu) < \infty$ and $I_t(\nu) < \infty$, then $S_{g\sharp}(\mu \times \nu) \ll \mathcal{L}^n$ for θ_n almost all $g \in O(n)$.*

Proof. For $\delta > 0$ let

$$\begin{aligned} A_\delta &= \{(u,v,x,y) \in (\mathbf{R}^n)^4 : \big||x-u| - |y-v|\big| \leq \delta \leq |x-u|/2\}, \\ B_\delta &= \{(u,v,x,y) \in (\mathbf{R}^n)^4 : |x-u| \leq 2\delta,\ |y-v| \leq 3\delta\}. \end{aligned}$$

Using Fatou's lemma, Theorem 1.19, Fubini's theorem and Lemma 3.8, we obtain

$$\begin{aligned} &\iint \liminf_{\delta\downarrow 0} \delta^{-n} S_{g\sharp}(\mu \times \nu)(B(z,\delta))\, dS_{g\sharp}(\mu\times\nu)z\, d\theta_n g \\ &\leq \liminf_{\delta\downarrow 0} \delta^{-n} \iint \mu \times \nu\big(\{(u,v) : |S_g(u,v) - S_g(x,y)| \leq \delta\}\big) \\ &\qquad \times d(\mu\times\nu)(x,y)\, d\theta_n g \\ &= \liminf_{\delta\downarrow 0} \delta^{-n} \iint \mu \times \nu\big(\{(u,v) : |x - u - g(y-v)| \leq \delta\}\big) \\ &\qquad \times d(\mu\times\nu)(x,y)\, d\theta_n g \\ &= \liminf_{\delta\downarrow 0} \delta^{-n} \int \theta_n\big(\{g : |x-u-g(y-v)| \leq \delta\}\big) \\ &\qquad \times d(\mu\times\nu\times\mu\times\nu)(u,v,x,y) \\ &\leq c_1 \liminf_{\delta\downarrow 0} \delta^{-1} \int_{A_\delta} |x-u|^{1-n}\, d(\mu\times\nu\times\mu\times\nu)(u,v,x,y) \\ &\qquad + \limsup_{\delta\downarrow 0} \delta^{-n}(\mu\times\nu\times\mu\times\nu)(B_\delta) \\ &= S + T, \end{aligned}$$

where S denotes the first summand and T the second. We estimate S using Fubini's theorem and Lemma 13.8:

$$\begin{aligned} S &= c_1 \liminf_{\delta\downarrow 0} \delta^{-1} \int_{\{(u,x):|x-u|\geq 2\delta\}} |x-u|^{1-n} \nu\times\nu\{(v,y) : \\ &\qquad \big||x-u| - |y-v|\big| \leq \delta\}\, d(\mu\times\mu)(u,x) \\ &\leq c_2 I_t(\nu) \iint |x-u|^{t-n}\, d\mu u\, d\mu x \\ &= c_2 I_{n-t}(\mu)\, I_t(\nu) < \infty \end{aligned}$$

because $n - t \leq s$, $I_s(\mu) < \infty$ and μ has compact support. To estimate T, observe that for $\delta > 0$

$$(2\delta)^{-s} \int \mu(B(x, 2\delta))\, d\mu x \leq \iint_{\{(u,x):|x-u|\leq 2\delta\}} |x-u|^{-s}\, d\mu u\, d\mu x,$$

which goes to zero as $\delta \downarrow 0$, since $I_s(\mu) < \infty$. Using Fubini's theorem and a similar estimate for ν we find

$$T \leq \limsup_{\delta\downarrow 0} \delta^{-s} \int \mu(B(x, 2\delta))\, d\mu x\, \delta^{-t} \int \nu(B(y, 3\delta))\, d\nu y = 0.$$

Hence $S + T < \infty$ and so for θ_n almost all $g \in O(n)$,

$$\liminf_{\delta\downarrow 0} \delta^{-n} S_{g\sharp}(\mu \times \nu)(B(z, \delta)) < \infty \quad \text{for } S_{g\sharp}(\mu \times \nu) \text{ almost all } z \in \mathbf{R}^n.$$

For any such g, Theorem 2.12 (3) gives $S_{g\sharp}(\mu \times \nu) \ll \mathcal{L}^n$. □

Next we prove an inequality for the energy-integrals. More details on measurabilities can be found in Mattila [4] and [7].

13.10. Lemma. *Suppose $0 < s < n$, $0 < t < n$, $s + t > n$ and $t \geq (n+1)/2$. If μ and ν are Radon measures on $\mathbf{R}^n$ with compact support and if $I_s(\mu) < \infty$ and $I_t(\nu) < \infty$, then*

$$\iint I_{s+t-n}(\mu \cap (\tau_z \circ g)_\sharp \nu)\, d\mathcal{L}^n z\, d\theta_n g \leq c I_s(\mu) I_t(\nu),$$

where c is a constant depending only on n and t.

Proof. Let $r = s + t - n$. Using (13.6), Fatou's lemma and Fubini's theorem, we have

$$\begin{aligned}
&\iint I_r(\mu \cap (\tau_z \circ g)_\sharp \nu)\, d\mathcal{L}^n z\, d\theta_n g \\
&= \iiiint |x-u|^{-r}\, d(\mu \cap (\tau_z \circ g)_\sharp \nu) x\, d(\mu \cap (\tau_z \circ g)_\sharp \nu) u\, d\mathcal{L}^n z\, d\theta_n g \\
&\leq \liminf_{\delta\downarrow 0} \alpha(n)^{-1} \delta^{-n} \iiiint_{W_{g,z}(\delta)} |x-u|^{-r} \\
&\qquad \times d(\mu \times \nu)(x, y)\, d(\mu \cap (\tau_z \circ g)_\sharp \nu) u\, d\mathcal{L}^n z\, d\theta_n g \\
&= \liminf_{\delta\downarrow 0} \alpha(n)^{-1} \delta^{-n} \iiint_{W_{g,z}(\delta)} \int |x-u|^{-r} \\
&\qquad \times d(\mu \cap (\tau_z \circ g)_\sharp \nu) u\, d(\mu \times \nu)(x, y)\, d\mathcal{L}^n z\, d\theta_n g.
\end{aligned}$$

Recalling that $W_{g,z}(\delta) = \{(x,y) : |x - g(y) - z| \leq \delta\}$, we use Fubini's theorem and (13.7) to get

$$\begin{aligned}
&\iint I_r(\mu \cap (\tau_z \circ g)_\sharp \nu)\, d\mathcal{L}^n z\, d\theta_n g \\
&\leq \liminf_{\delta \downarrow 0} \alpha(n)^{-1} \delta^{-n} \iiint_{\{z : |x-g(y)-z| \leq \delta\}} \int |x - u|^{-r} \\
&\qquad \times d(\mu \cap (\tau_z \circ g)_\sharp \nu) u\, d\mathcal{L}^n z\, d(\mu \times \nu)(x,y)\, d\theta_n g \\
&\leq \liminf_{\delta \downarrow 0} \alpha(n)^{-1} \delta^{-n} \iiint_{\{(u,v) : |x-g(y)-(u-g(v))| \leq \delta\}} |x - u|^{-r} \\
&\qquad \times d(\mu \times \nu)(u,v)\, d(\mu \times \nu)(x,y)\, d\theta_n g \\
&= \liminf_{\delta \downarrow 0} \alpha(n)^{-1} \delta^{-n} \int \theta_n\big(\{g : |x - u - g(y - v)| \leq \delta\}\big) |x - u|^{-r} \\
&\qquad \times d(\mu \times \nu \times \mu \times \nu)(u,v,x,y).
\end{aligned}$$

Defining the sets A_δ and B_δ as in the proof of Lemma 13.9 and applying Lemma 3.8, we obtain

$$\begin{aligned}
&\iint I_r(\mu \cap (\tau_z \circ g)_\sharp \nu)\, d\mathcal{L}^n z\, d\theta_n g \\
&\leq c_1 \liminf_{\delta \downarrow 0} \delta^{-1} \int_{A_\delta} |x - u|^{1-s-t}\, d(\mu \times \nu \times \mu \times \nu)(u,v,x,y) \\
&\quad + \alpha(n)^{-1} \limsup_{\delta \downarrow 0} \delta^{-n} \int_{B_\delta} |x - u|^{-r}\, d(\mu \times \nu \times \mu \times \nu)(u,v,x,y).
\end{aligned}$$

As in the proof of Lemma 13.9 one finds that the first summand is bounded by $cI_s(\mu)I_t(\nu)$, with c depending only on n and t, and the second is zero. □

Hausdorff dimension and capacities of intersections

Now we are ready to prove a result on the dimension of intersections.

13.11. Theorem. *Let $s, t > 0$, $s + t > n$ and $t > (n+1)/2$. If A and B are Borel sets in $\mathbf{R}^n$ with $\mathcal{H}^s(A) > 0$ and $\mathcal{H}^t(B) > 0$, then for θ_n almost all $g \in O(n)$,*

$$\mathcal{L}^n\big(\{z \in \mathbf{R}^n : \dim(A \cap (\tau_z \circ g)B) \geq s + t - n\}\big) > 0.$$

Proof. By Frostman's lemma 8.8 there are Radon measures μ and ν with compact support such that $\operatorname{spt} \mu \subset A$, $\operatorname{spt} \nu \subset B$, $\mu(A) > 0$, $\nu(B) > 0$,

$\mu(B(x,r)) \le r^s$ and $\nu(B(x,r)) \le r^t$ for $x \in \mathbf{R}^n$, $r > 0$. Then (as in Chapter 8), $I_p(\mu) < \infty$ for $0 < p < s$ and $I_q(\nu) < \infty$ for $0 < q < t$. When in addition $p + q > n$ and $(n+1)/2 < q < t$ we have by Lemma 13.10 for θ_n almost all $g \in O(n)$,

$$I_{p+q-n}\big(\mu \cap (\tau_z \circ g)_\sharp \nu\big) < \infty \quad \text{for } \mathcal{L}^n \text{ almost all } z \in \mathbf{R}^n.$$

Using Lemma 13.9 and (13.7), we have

$$\int \big(\mu \cap (\tau_z \circ g)_\sharp \nu\big)(\mathbf{R}^n)\, d\mathcal{L}^n z = \mu(\mathbf{R}^n)\,\nu(\mathbf{R}^n) > 0$$

for θ_n almost all $g \in O(n)$, whence

$$\mathcal{L}^n(E_g) > 0 \quad \text{where } E_g = \big\{z \in \mathbf{R}^n : \big(\mu \cap (\tau_z \circ g)_\sharp \nu\big)(\mathbf{R}^n) > 0\big\}.$$

Recalling (13.5) we see that $C_{p+q-n}(A \cap (\tau_z \circ g)B) > 0$ for $\mathcal{L}^n$ almost all $z \in E_g$ and θ_n almost all $g \in O(n)$. This gives $\dim A \cap (\tau_z \circ g)B \ge p+q-n$ by Theorem 8.9 (1). The theorem follows now letting $p \uparrow s$ and $q \uparrow t$ (note that E_g is independent of p and q). □

The opposite inequality does not hold in this generality, see Example 13.19 and Falconer [7]. However, we shall now show that it holds if A and B obey the product rule $\dim(A \times B) = \dim A + \dim B$. Recall from Corollary 8.11 that this holds if the Hausdorff and packing dimensions of B agree; in particular, if $0 < \mathcal{H}^t(B) < \infty$ and $\Theta^t_*(B, y) > 0$ for $y \in B$ by Theorem 6.13.

13.12. Theorem. *Let A and B be Borel sets in $\mathbf{R}^n$. If $\dim(A \times B) \ge n$, then*

$$\dim(A \cap \tau_z(B)) \le \dim(A \times B) - n$$

for $\mathcal{L}^n$ almost all $z \in \mathbf{R}^n$.

Proof. As observed in (13.2)

$$A \cap \tau_z(B) = \pi((A \times B) \cap V_z),$$

where $\pi(x,y) = x$ and $V_z = \{(x,y) : x - y = z\}$. Obviously, $|\pi a - \pi b| = |a - b|/\sqrt{2}$ for $a, b \in V_z$, whence for any $s > 0$,

$$\mathcal{H}^s(A \cap \tau_z(B)) = 2^{-s/2}\mathcal{H}^s((A \times B) \cap V_z).$$

Now $V_z = S^{-1}\{z\}$ for the map $S\colon \mathbf{R}^n \times \mathbf{R}^n \to \mathbf{R}^n$ defined by $S(x,y) = x - y$. Hence by Theorem 7.7, with c depending on n and t,

$$\int^* \mathcal{H}^{t-n}(A \cap \tau_z(B))\, d\mathcal{L}^n z \le c\mathcal{H}^t(A \times B)$$

for any $n \le t \le 2n$. This gives immediately the desired inequality. □

Putting together the last two theorems we obtain

13.13. Corollary. *Let s, $t > 0$, $s + t \geq n$ and $t > (n+1)/2$. If A and B are Borel sets in $\mathbf{R}^n$ with $\dim A = s$, $\mathcal{H}^s(A) > 0$, $\dim B = t$ and $\mathcal{H}^t(B) > 0$, and if $\dim(A \times B) = s + t$, then for θ_n almost all $g \in O(n)$,*

$$\mathcal{L}^n\big(\{z \in \mathbf{R}^n : \dim(A \cap (\tau_z \circ g)B) = s + t - n\}\big) > 0.$$

The above methods can be used also to get other variants of these results. We state two. Their proofs can be found in Mattila [7] and [9].

13.14. Theorem. *Let s, $t > 0$, $s + t \geq n$ and $t > (n+1)/2$. If A is $\mathcal{H}^s$ measurable with $\mathcal{H}^s(A) < \infty$ and B is $\mathcal{H}^t$ measurable with $\mathcal{H}^t(B) < \infty$, then for $\mathcal{H}^s \times \mathcal{H}^t \times \theta_n$ almost all $(x, y, g) \in A \times B \times O(n)$,*

$$\dim A \cap (\tau_x \circ g \circ \tau_{-y})\, B \geq s + t - n.$$

This holds as equality if in addition,

$$\dim(A \times B) = \dim A + \dim B.$$

Note that the map $\tau_x \circ g \circ \tau_{-y}$ is composed of a rotation around y and a translation sending y to x.

13.15. Theorem. *If s, $t > 0$, $s + t > n$ and $t \geq (n+1)/2$, then for any A, $B \subset \mathbf{R}^n$*

$$C_s(A)\, C_t(B) \leq c \int_* \int_* C_{s+t-n}(A \cap (\tau_z \circ g)B)\, d\mathcal{L}^n z\, d\theta_n g$$

where c is a constant depending only on n, s and t.

13.16. Remarks. As was already noted it is not known in the above theorems whether one of the dimensions must be at least $(n+1)/2$, when $n \geq 2$ (see 13.18 for $n = 1$). Since their sum can always be assumed to be at least n, one of them is then at least $n/2$ and so there is a gap of $1/2$ of uncertainty. Mattila [13] developed the Fourier-analytic methods further to give partial results for general dimensions. For example, it was shown that if one of the sets A and B is a Salem set (recall 12.17 for the definition), then no extra assumption on the dimension is needed.

If one considers larger transformation groups in place of isometries no such assumption is needed either, see Kahane [4] and Mattila [7]. For example if one takes the group generated by the orthogonal group and the homotheties $\delta_r \colon \mathbf{R}^n \to \mathbf{R}^n$, $\delta_r(x) = rx$, $x \in \mathbf{R}^n$, $0 < r < \infty$, analogues of all the results 13.11–15 hold for general dimensions. Here n can also be 1. The resulting maps are then the similarity maps $f\colon \mathbf{R}^n \to \mathbf{R}^n$ which change the distance by a constant factor, $|f(x)-f(y)| = r|x-y|$. Then also an integration with respect to r is involved, which makes things easier. The use of the Fourier transform can be avoided in this case. More generally, Kahane [4] showed that one can use general subgroups of all linear bijections $\mathbf{R}^n \to \mathbf{R}^n$. For example, the following theorem can be found in Kahane [4].

13.17. Theorem. *Let G be a closed subgroup of the general linear group of $\mathbf{R}^n$ which is transitive in $\mathbf{R}^n \setminus \{0\}$ and let τ be a Haar measure of G. If $s+t>n$ and A and B are Borel sets in $\mathbf{R}^n$ with $C_s(A)>0$ and $C_t(B)>0$, then for τ almost all $g \in G$,*

$$\mathcal{L}^n\big(\{z \in \mathbf{R}^n : C_{s+t-n}(A \cap (\tau_z \circ g)B) > 0\}\big) > 0.$$

Examples and remarks

We now discuss some examples. The assumption $t > (n+1)/2$ makes the above results rather empty if $n=1$, and the following examples show that there is really nothing to say in this respect.

13.18. Example. There are compact subsets A and B of $\mathbf{R}$ such that $\dim A = \dim B = 1$ and $A \cap \tau_z(B)$ contains at most one point for every $z \in \mathbf{R}$.

Let $0<s<1$. We just indicate the idea of how to construct compact sets A and B of dimension s such that every translate of B contains at most one point of A. We use the very porous Cantor sets discussed at the end of 4.12. Both A and B will be Cantor sets of the form

$$A = \bigcap_{k=1}^{\infty} \bigcup_{i_1\cdots i_k} I_{i_1\cdots i_k}, \qquad B = \bigcap_{k=1}^{\infty} \bigcup_{i_1\cdots i_k} J_{i_1\cdots i_k},$$

constructed on $I_1 = J_1 = [0,1]$. All the intervals $I_{i_1\cdots i_k}$ and $J_{i_1\cdots i_k}$ have the same length d_k which satisfies

$$n_{k+1} d_{k+1}^s = d_k^s, \quad d_1 = 1.$$

Here n_{k+1} is the number of the intervals $I_{i_1,\ldots,i_k,i}$, $i=1,\ldots,n_{k+1}$, inside each $I_{i_1,\ldots,i_k}$ and also the number of the intervals $J_{i_1,\ldots,i_k,i}$ inside $J_{i_1,\ldots,i_k}$. We achieve what we want by distributing the intervals $I_{i_1,\ldots,i_k,i}$ inside

$I_{i_1,\dots,i_k}$ somewhat differently from the intervals $J_{i_1,\dots,i_k,i}$ inside $J_{i_1,\dots,i_k}$. The I-intervals are distributed with equal distances c_k starting from the left end-point and leaving a gap of c_k before the right end-point. For the J-intervals we use the following trick. If $i_k = 1$ place the intervals $J_{i_1,\dots,i_k,i}$ inside $J_{i_1,\dots,i_k}$ as in the case of $I_{i_1,\dots,i_k}$ starting from the left end-point. For $i_k > 1$ translate the intervals to the right by $3(i_k - 1)d_{k+1}$ so that you start from $a + 3(i_k - 1)d_{k+1}$ where a is the left end-point of $J_{i,\dots,i_k}$. Suppose d_k decreases so quickly that $4n_k d_{k+1} < c_k$. Then it follows from this construction that for any $z \in \mathbf{R}$, the intersection

$$\Big(\bigcup_{i=1}^{n_{k+1}} I_{i_1,\dots,i_k,i}\Big) \cap \tau_z\Big(\bigcup_{j=1}^{n_{k+1}} J_{j_1,\dots,j_k,j}\Big)$$

can be non-empty for at most one pair of sequences $i_1,\dots,i_k$ and $j_1,\dots,j_k$. This yields that $A \cap \tau_z(B)$ is either empty or a singleton.

We leave the details, some of which are given in Mattila [7], to the reader as well as the modification to the case $s = 1$.

Examples in the opposite direction can be given in any $\mathbf{R}^n$. The following and more were proven by Falconer [7], see also Dodson, Rynne and Vickers [2], Falconer [16, 8.2 and 10.3], [24] and Rynne [1]; these references contain discussions on relations to Diophantine approximation. The case $n = 1$ can be found in Mattila [7].

13.19. Example. For any $0 \le s \le n$ there exists a Borel set A in $\mathbf{R}^n$ such that $\dim A = s$ and $\dim A \cap f(A) = s$ for all similarity maps $f\colon \mathbf{R}^n \to \mathbf{R}^n$.

To get an idea how this might happen just consider one finite union of subintervals of $I = [0,1]$. Let $0 < t < s < 1$. Choose n_1 uniformly distributed intervals $I_{1,1},\dots,I_{1,n_1} \subset I$ of length d_1. Next choose n_2 roughly uniformly distributed intervals in $I \setminus \bigcup_{i=1}^{n_1} I_{1,i}$ of length $d_2 < d_1$. Continue this m times. All this can be arranged so that the lengths $d_1,\dots,d_m$ decrease rapidly and satisfy

$$\sum_{k=1}^{m} n_k\, d_k^s = 1 \quad \text{and} \quad n_k\, d_k^t \ge M \quad \text{for } k = 1,\dots,m,$$

where M is a large number. Thus we have taken a step in a construction of a set of dimension s. If m is large, the intervals $I_{k,i}$ will be rather dense in I and one can see that if $z \in [0,1/2]$ and the intersection $(\bigcup_{i,k} I_{k,i}) \cap \tau_z(\bigcup_{i,k} I_{k,i})$ is covered by intervals J_i, then $\sum_i d(J_i)^t \ge 1$.

When we go on with the construction t can be allowed to vary and approach s guaranteeing that the intersections will have dimension s.

13.20. Remarks. As shown in Example 13.18 the conclusion of Theorem 13.11 does not hold in $\mathbf{R}^1$; in fact it fails even for $A = B = C(1/3)$ where $C(1/3)$ is the Cantor ternary set of 4.10. This follows from a result of Hawkes [2] who proved that

$$\dim C(1/3) \cap (C(1/3) + z) = \log 2/(3 \log 3)$$

for $\mathcal{L}^1$ almost all $z \in (0,1)$. Kenyon and Peres [1] gave extensions of this to more general Cantor sets. For some other questions on intersections of Cantor sets, see Hunt, Kan and Yorke [1] and Kraft [1]. The motivation for studying Cantor sets in this and many other respects comes from dynamical systems.

Kahane [4], see also Falconer [16, § 16.1], applied intersection results to study the multiple points, i.e. points visited more than once, and related questions of Brownian motion and other stochastic processes. Briefly the idea is the following. Let E and F be compact subsets of $\mathbf{R}$ lying in disjoint closed intervals. Then with positive probability for the Brownian motion $\omega\colon [0,\infty) \to \mathbf{R}^n$, $n \geq 2$, recall Theorem 9.18,

$$\begin{aligned} \dim \big(\omega(E) \cap \omega(F)\big) &= \max \big\{ \dim \omega(E) + \dim \omega(F) - n, 0 \big\} \\ &= \max\{2 \dim E + 2 \dim F - n, 0\}. \end{aligned}$$

Intersection results can be used to prove this since, due to the basic properties of the Brownian motion, $\omega(F)$ and $f(\omega(F))$ are sufficiently closely related random sets for any similarity map $f\colon \mathbf{R}^n \to \mathbf{R}^n$. For example, if $n = 3$ and E and F are intervals, it follows that $\omega(E) \cap \omega(F) \neq \emptyset$ with positive probability, which leads to the well-known fact that the 3-dimensional Brownian motion has almost surely double points.

Falconer [19] gave an application of the intersection results on a combinatorial problem on Hausdorff dimension. Hawkes [1] studied the dimension rule for the intersections from the point of view of independence.

Exercises.

1. Let $W \in G(n,m)$. Prove that

$$\dim(V \cap W) = \max\{k + m - n, 0\}$$

for $\gamma_{n,k}$ almost all $V \in G(n,k)$.

2. Show that $A \cap \tau_z(B) = \emptyset$ for $\mathcal{L}^n$ almost all $z \in \mathbf{R}^n$ if A and B are Borel subsets of $\mathbf{R}^n$ with $\dim(A \times B) < n$.
3. Let A, B and C be Borel subsets of $\mathbf{R}^n$, $n \geq 2$, such that $\dim A = \dim B = \dim C = s$ and $s > \max\{(n+1)/2, 2n/3\}$. Show that there exist isometries $f_1, f_2 \colon \mathbf{R}^n \to \mathbf{R}^n$ such that $A \cap f_1(B) \cap f_2(C) \neq \emptyset$.